

111

Ex. 1

* Integration by Partial fraction.

If we want integrate $\frac{P(x)}{Q(x)}$ ← are polynomials of x . First we decomposed $\frac{P(x)}{Q(x)}$ before carrying out the process of integration.

Ex. 1: Find $\int \frac{2x+1}{x^2+2x-3} dx$

$$\therefore \frac{2x+1}{x^2+2x-3} \stackrel{\text{sol.}}{=} \frac{2x+1}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}$$

$$\Rightarrow 2x+1 = A(x+3) + B(x-1)$$

$$\text{at } x=1 \Rightarrow 3 = 4A \Rightarrow A = \frac{3}{4}$$

$$\text{at } x=-3 \Rightarrow -5 = -4B \Rightarrow B = \frac{5}{4}$$

$$\therefore \frac{2x+1}{x^2+2x-3} = \frac{\frac{3}{4}}{x-1} + \frac{\frac{5}{4}}{x+3}$$

$$\begin{aligned} \Rightarrow \int \frac{2x+1}{x^2+2x-3} dx &= \frac{3}{4} \int \frac{1}{x-1} dx + \frac{5}{4} \int \frac{1}{x+3} dx \\ &= \frac{3}{4} \ln(x-1) + \frac{5}{4} \ln(x+3) + C \end{aligned}$$

Ex: Evaluate $\int \frac{x^3 - 2x}{x^2 + 3x + 2} dx$

Sol:

Since the degree of numerator is greater than degree of denominator, we use the long division first

to obtain

$$\int \frac{x^3 - 2x}{x^2 + 3x + 2} dx = \int \left[x - 3 + \frac{5x + 6}{x^2 + 3x + 2} \right] dx$$

$$= \int \left[x - 3 + \frac{1}{x+1} + \frac{4}{x+2} \right] dx$$

$$= \frac{x^2}{2} - 3x + \ln(x+1) + 4\ln(x+2) + C$$

Ex. 3 Find $\int \frac{x^2 + 2x + 4}{(x+1)^3} dx$

Sol:

Decomposed the integrand into partial fraction

$$\frac{x^2 + 2x + 4}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$$

$$\Rightarrow x^2 + 2x + 4 = (x+1)^2 A + (x+1) B + C$$

$$= Ax^2 + 2Ax + A + Bx + B + C$$

$$\Rightarrow A = 1 \quad \& \quad 2A + B = 2 \Rightarrow B = 0$$

$$A + B + C = 4 \Rightarrow C = 3$$

$$\Rightarrow \int \frac{x^2 + 2x + 4}{(x+1)^3} dx = \int \left(\frac{1}{x+1} + \frac{3}{(x+1)^3} \right) dx$$

$$= \ln(x+1) - \frac{3}{2} (x+1)^{-2} + C$$

=

Evaluate $\int \frac{20x}{(x^2+1)(x^2+2x+5)} dx$ ③

1. Decomposed The integrand into partial fractions

Let $\frac{20x}{(x^2+1)(x^2+2x+5)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+2x+5}$

$\Rightarrow 20x = (Ax+B)(x^2+2x+5) + (Cx+D)(x^2+1)$

$\therefore 20x = \cancel{Ax^3} + 2Ax^2 + \cancel{Bx^2} + 5Ax + \cancel{2x^3} + \cancel{Cx^3} + \cancel{Cx^2} + \cancel{Dx^2} + \cancel{D}$

$\Rightarrow A+C=0 \Rightarrow A=-C$

$2A+B+D=0 \Rightarrow 2A+B-5B=0 \Rightarrow A=2B$

$5A+2B+C=20 \Rightarrow 5A+A-A=20 \Rightarrow A=4$

$5B+D=0 \Rightarrow D=-5B$

Then $A=4$, $B=2$, $C=-4$ and $D=-10$

$\Rightarrow \int \frac{20x}{(x^2+1)(x^2+2x+5)} dx = \int \left[\frac{4x+2}{x^2+1} - \frac{4x+10}{x^2+2x+5} \right] dx$

$= 2 \ln(x^2+1) + 2 \tan^{-1} x - 2 \ln(x^2+2x+5) - \tan^{-1} \frac{x+1}{2} + C$

2. $\frac{4x+2}{x^2+1} = 2 \frac{2x}{x^2+1} + \frac{2}{x^2+1}$
 $\frac{4x+10}{x^2+2x+5} = \frac{2(x+1)}{(x+1)^2+4} = \frac{2(x+1)}{(x+1)^2+4} + \frac{6}{(x+1)^2+4}$

blems: Evaluate the following integrals

① $\int \frac{6x-1}{x^2(2x-1)} dx$

② $\int \frac{dx}{x^2-3x}$

③ $\int \frac{dx}{2x^2+6x}$

④ $\int \frac{x+2}{2x^2+x} dx$

⑤ $\int \frac{2x-1}{x^3+2x^2} dx$

⑥ $\int \frac{dx}{x^2+x-2}$

⑦ $\int \frac{x dx}{(x^2-1)^2}$

⑧ $\int \frac{x^2 dx}{(x^2+4)(x-1)^3}$

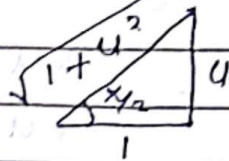
=

①

* substituting by half angle:

Using when The integral contains
 $\sin x$ & $\cos x$ & $\tan x$
 in this case take

$$u = \tan \frac{x}{2}$$



$$\Rightarrow du = \frac{1}{2} \sec^2 \frac{x}{2} dx \quad \text{also } 2 \sin \frac{x}{2} \cos \frac{x}{2} = \sin x$$

$$\sin x = \frac{2u}{1+u^2}$$

$$\begin{cases} \cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \\ \tan x = \frac{2 \tan \frac{x}{2}}{1 - \tan^2 \frac{x}{2}} \end{cases}$$

$$\cos x = \frac{1-u^2}{1+u^2}$$

$$\tan x = \frac{2u}{1-u^2}$$

Ex: Evaluate the following integrals:

① $\int \frac{dx}{2 + \cos x}$

② $\int \frac{dx}{\sin x + \tan x}$

Solution

① let $u = \tan \frac{x}{2} \Rightarrow du = \frac{1}{2} \sec^2 \frac{x}{2} dx$

$$\Rightarrow dx = \frac{2}{\sec^2 \frac{x}{2}} du = \frac{2}{1+u^2} du$$

$$\Rightarrow \int \frac{dx}{2 + \cos x} = \int \frac{\cancel{dx} \cdot 2}{1+u^2} du$$

$$2 + \frac{1-u^2}{1+u^2}$$

$$= \int \frac{2}{2 + 2u^2 + 1 - u^2} du$$

$$= \int \frac{2}{3 + u^2} du = \frac{2}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} + c$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \tan \frac{x}{2} \right) + c$$

$$\textcircled{2} I = \int \frac{dx}{\sin x + \tan x}$$

$$u = \tan \frac{x}{2} \Rightarrow dx = \frac{2}{1+u^2} du$$

$$\sin x = \frac{2u}{1+u^2}, \quad \tan x = \frac{2u}{1-u^2}$$

$$I = \int \frac{\frac{2}{1+u^2} du}{\frac{2u}{1+u^2} + \frac{2u}{1-u^2}}$$

$$= \int \frac{2(1-u^2)}{u-u^3+u+u^3} du$$

$$I = \int \frac{1-u^2}{2u} du$$

$$I = \frac{1}{2} \ln u - \frac{1}{2} u^2 + C$$

$$\Rightarrow I = \frac{1}{2} \ln \left(\tan \frac{x}{2} \right) - \frac{1}{2} \left(\tan \frac{x}{2} \right)^2 + C$$

Problems:

Find the following integrals.

$$\textcircled{1} \int \frac{dx}{1+4\sin x}$$

$$\textcircled{2} \int \frac{dx}{5+4\cos x - \sin x}$$

$$\textcircled{3} \int \frac{dx}{5+4\cos x}$$

$$\textcircled{4} \int \frac{dx}{1-3\sin x}$$

(3)

دفعات التوافق / ٩

$$\sin mx \sin nx$$

$$= \frac{1}{2} [\cos(m-n)x - \cos(m+n)x]$$

$$\cos mx \cos nx = \frac{1}{2} [\cos(m-n)x + \cos(m+n)x]$$

$$\sin mx \cos nx = \frac{1}{2} [\sin(m-n)x + \sin(m+n)x]$$

Ex.

$$\text{Evaluate } \int \sin 3x \cos 5x \, dx$$